From CM1/WRF Thunderstorm Downburst Wind Fields to Aerodynamic Loads on High-Rise Buildings: An Executable Cross-Scale CFD Workflow, Validation Ladder, and Evidence Boundaries

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Scientific approval status. `scientific_approval_status = not_scientifically_approved`. This review is at the internal scientific review stage and does not constitute engineering adoption, code provisions, solver selection, or final load conclusions. Literature excerpt figures are labelled "internal-review excerpt"; copyright and permissions must be verified figure by figure before external publication. Evidence-status markers: [reported] source-reported value, [derived] derived value, [bounded] bounded interpretation, [not reported] not reported. Total corpus of 68 full-text sources: 57 retained, 11 excluded. Excluded sources do not appear in the body text, tables, or references. Independent experimental directed-validation edge ledger: one entry. That edge supports only source-limited CFD pressure or drag comparison; no single source validates the complete CM1/WRF-to-high-rise-load chain.

Abstract

Transforming three-dimensional transient thunderstorm downburst wind fields produced by CM1 or WRF into aerodynamic loads on high-rise buildings requires establishing continuous and auditable interfaces among source-field generation, variable contract and coordinate transformation, spatial and temporal interpolation, interface turbulence reconstruction, building-resolving CFD solution, pressure reference and coefficient normalisation, and load integration. This review adopts a fixed target of an isolated, rigid, sealed, sharp-edged square prism with parameters H=100 m, B=D=15 m, normal incidence, 25 facade bands, and no internal pressure. The 57 retained sources are organised by their relationships as input packages, transformations or solutions, quality-assurance checks, and output packages, rather than listed source by source.

On the source-field side, full-cloud CM1 preserves microphysics-driven outflow asymmetry; cold-pool models can approach full-cloud results in peak amplitude while the peak-radius position still differs by 22.2% to 32.0%, and the vorticity discrepancy between impinging jets and cold pools reaches 55.9% to 58.0%. On cross-scale transfer, only two sources directly demonstrate CM1-to-second-solver velocity-field transfer; 19 WRF-to-microscale interface analogues cover a wide range of schemes from static profiles to fully dynamic driving, yet none validates the error from convective-downburst to building-pressure. On building aerodynamics, ABL high-rise LES mean-pressure R^2 can exceed 0.90, base-moment discrepancies are 10.5% to 13.0%, and local-extreme coverage is 77.6% to 90.4%. Accelerating-flow cross-section studies report peak-drag exceedances of 22.4% to 109.7%. A single-GPU case with approximately 9 M cells can be completed within 6 h, whereas multi-directional high-rise LES budgets can reach tens of thousands to hundreds of thousands of core-hours.

This evidence supports the construction of an eight-rung validation ladder, three bounded candidate routes, and a link-by-link resource budget. Neural operators, reduced-order models,

synthetic turbulence, and adaptive mesh refinement enter candidate routes only when they can preserve the target observable. All numerical values are constrained by their respective geometry, flow type, normalisation, grid, and validation object and cannot be spliced into a single validated end-to-end chain.

Keywords: downburst; CM1; WRF; high-rise building aerodynamic loads; cross-scale CFD workflow; validation ladder; evidence boundaries

1 Introduction and precise objective chain

A downburst is an intense transient wind field formed when a downdraft within a thunderstorm system reaches the ground and diverges radially outward. Its near-surface velocity profile, duration, spatial structure, and unsteady characteristics differ fundamentally from those of a stationary atmospheric boundary layer wind. When such a transient wind field acts on a high-rise building, the extraction of aerodynamic endpoints including facade pressure, storey forces, base shear, overturning moments, and torsion constitutes a multiphysics problem spanning meteorological to engineering scales.

This review focuses on an operational question: given three-dimensional transient downburst output produced by CM1 or WRF, how to construct a reproducible one-way multiscale workflow that sequentially establishes fine-scale CFD boundary and initial conditions, resolves the wind field around a high-rise building, and extracts facade pressure together with 25-band storey forces, base shear, two overturning moments M_x and M_y, and torsion M_z. The fixed target is an isolated, rigid, sealed, sharp-edged square-prism high-rise building with parameters H=100 m, B=D=15 m (aspect ratio approximately 6.7), normal incidence, 25 facade-band definition, and no internal pressure. Dynamic structural response is explicitly excluded from the endpoints.

The complete chain can be decomposed into seven links (Figure 1): L0 hazard source-field generation, L1 boundary-variable and coordinate contract, L2 spatial and temporal interpolation, L3 interface turbulence reconstruction, L4 building-scale solver and near-wall grid, L5 pressure reference and coefficient normalisation, L6 load integration. A deterministic QA gate is placed between each pair of adjacent links: input package $\rightarrow$ transformation or solution action $\rightarrow$ QA gate $\rightarrow$ output package $\rightarrow$ validation status. This executable-interface central interpretive unit runs through the entire text.

The 57 retained sources cover several necessary links of the chain, but no single source simultaneously possesses a high-fidelity downburst source field, an explicit CM1-to-CFD variable contract, H=100 m target geometry, local pressure measurements, and reproducible computational cost. Evidence for end-to-end CM1 or WRF downburst to target-building aerodynamic loads has therefore not yet closed. The eligibility audit further excluded 11 sources containing no CFD contribution or lacking a necessary validation relationship. Experimental data enter through 10 same-paper experiment-plus-CFD studies, each limited to its actually compared observable range; one independent experiment enters through a single verified directed relationship used for the source-limited comparison in Section 7.

Six interface fractures prevent an end-to-end validation claim: (1) the variable contract between CM1 or WRF source fields and fine-scale CFD is incomplete; (2) the influence of source-field output frequency and interpolation on peak position has not been quantified on the same downburst event;

(3) interface turbulence reconstruction has not been validated under non-stationary downburst conditions; (4) building-scale grid convergence of mean pressure, fluctuating pressure, local peaks, and moments has not been simultaneously reported; (5) pressure-to-moment integration amplification has not been checked under downburst conditions; (6) data-conversion, inflow-development, and post-processing costs are typically omitted from end-to-end resource budgets.

The following sections are organised not as source-by-source summaries but along the logic of executable interfaces: Section 2 defines what CM1 and WRF must export; Section 3 describes the source-field preparation sequence; Section 4 compares six fine-scale CFD boundary and initial-condition construction strategies; Section 5 discusses fine-scale wind-field CFD; Section 6 defines high-rise building pressure and load extraction; Section 7 constructs the eight-rung validation ladder; Section 8 proposes three bounded candidate workflows; Section 9 compresses remaining research needs; Section 10 provides concluding remarks.

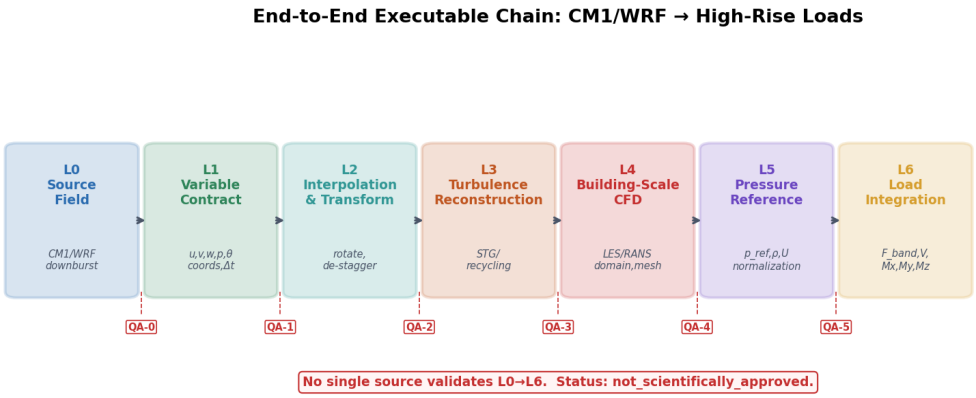


Figure 1: Schematic decomposition of the end-to-end executable chain L0–L6 and QA gates. No single source closes the full L0-to-L6 chain. This figure is a project-original synthesis diagram and does not constitute validation evidence.

2 What CM1/WRF must export

The completeness of source-field output variables determines which transformations the downstream interface can execute and which physical consistencies can only be checked rather than recovered. The variable contract is therefore divided into three tiers: minimum required, recommended, and solver-specific (Table 2); the higher the tier, the more complete the thermodynamic, turbulence, and boundary consistencies that can be checked, but one cannot infer from a tier that a given source actually output unreported variables.

2.1 Minimum required variables

The minimum required contract comprises three-dimensional velocity components u, v, w; a timestamp sequence; grid metrics (horizontal spacing and vertical-layer definition); coordinate-system definition (origin, rotation, projection parameters); terrain or surface elevation; vertical-layer semantics (the distinction among height layers, mass layers, or flux layers and the choice of σ or z coordinate); and source identification with provenance information (model version, configuration,

Table 1: Source eligibility and exclusion summary. The 57 retained sources constitute the citation ceiling; the 11 excluded sources must not appear in the body text, tables, or references.

Category	**Count**	**Description**
Full-text corpus total	68	Auditable sources that completed full-text eligibility review
Retained sources	**57**	
CM1/WRF source-field simulations*	16	Full-cloud CM1, cold-pool parameterisations, and related source fields
Mesoscale-to-microscale coupling*	23	Dynamic driving, profile transfer, synthetic turbulence, etc.
Fine-scale CFD*	40	LES, RANS, URANS, IBM, and related numerical methods
Building pressure or load CFD*	18	Pressure, force, moment, or extraction methodology
Same-paper experiment plus CFD*	10	Supporting only the observables actually compared in each paper
Implementation reviews or method sources*	5	Supporting method classification; no primary performance conclusions
Independent experimental directed-validation source	1	Entering Section 7 only through the VE-001 validation edge
Excluded sources	**11**	
Independent experiments (no CFD)	8	Wind-tunnel or jet tests with no retained CFD validation edge
Structural/analytical models	3	Assumed aerodynamic coefficients; no flow-field CFD
Independent experimental directed-validation edge	1	VE-001: experimental source used only for source-limited validation of a retained CFD source
Same-paper experiment + CFD validation edges	10	Valid only within their respective observable ranges

*Source categories may overlap; counts indicate category membership and should not be summed to the retained total.

and initial-condition identifier). The absence of any one of these prevents downstream coordinate transformation or interpolation from being executed or audited.

The CM1 source-field family shows markedly different reporting coverage for these variables. Orf et al. (2012) simulated a thunderstorm that produced a downburst, with asymmetric outflow height and a brief near-surface peak[reported]. Saville et al. (2014) applied circumferential analysis to the same simulation and reported a peak-to-mean ratio of 1.936[reported], conditioned on the three strongest wind instants and the radial range not dominated by an adjacent downdraft. Orf et al. (2017) simulated a long-track tornadic supercell in CM1 version 16 with a 30 m inner grid and 0.05 s time step[reported]. Orf et al. (2019)'s quarter-trillion-cell simulation used the LOFS framework to support 0.2 s subdomain high-frequency output[reported]. These sources demonstrate that CM1 can produce high-spatiotemporal-resolution three-dimensional transient wind fields, but the complete thermodynamic and turbulence variable contract is [not reported] in most sources.

2.2 Recommended variables

The recommended contract adds, beyond the minimum required, pressure or pressure perturbation (distinguishing full from perturbation and reporting the reference state), potential temperature θ, temperature T, moisture variables q_v, and density ρ or a thermodynamic variable combination sufficient to reconstruct density. These variables support thermodynamic consistency checks and downstream CFD density, pressure, and buoyancy initialisation.

Vermeire (2016) conducted a cold-pool parameterisation study in CM1 with a 9.6 km domain, 10 m horizontal grid, 0.05 s time step, and 147 M grid points[reported]; an exponential relationship between peak cooling rate and peak radial wind speed was found within the ten-case matrix[reported]. Nguyen (2021) systematically studied cold-pool scaling in cm1r19.10; in the four fixed-geometry cases of Chapter 3, the peak radial wind speed at t/T_0=4.0 scaled as $u_{r,\max}/V_0$=3.25, with RMS deviation below 10%[reported]. These studies reported velocity and partial thermodynamic variables, but the complete pressure reference state, density output, and compression strategy remain [not reported] in most configurations.

2.3 Solver-specific variables

Solver-specific variables include TKE or SGS variables (supporting downstream inflow-turbulence initialisation), staggering metadata (Arakawa C or B grid position specification), surface fields (fluxes, roughness), restart identifiers and checksums (one checksum per event subdomain per snapshot to support reversible auditing), and boundary-condition provenance (lateral-boundary and top-driving source records).

Vermeire et al. (2011) reported independent CM1 cold-pool and impinging-jet configurations; the maximum deviation of four non-vorticity parameters for the designated case pairs was 6.5% and 5.1%, whereas the non-dimensional vorticity deviation reached 55.9% and 58.0%[reported], indicating that impinging jets overestimate velocity components above the maximum radial-wind-speed height and do not capture the vorticity structure of cold pools. Orf et al. (2011)'s downburst-line study found that interacting downbursts within the tested two-event matrix increased horizontal peak wind speed[reported]. These sources did not fully report solver-specific variables such as staggering metadata and boundary-condition provenance.

2.4 Event export requirements

Event export requires simultaneous recording of source-domain boundaries, selected event time window, storm track and translation velocity, source-grid output frequency Δt_{out}, subdomain margin, output precision, units, missing-value convention, and one checksum per snapshot. Anabor et al. (2018) compared full-cloud CM1 with axisymmetric URANS release, reporting that peak radial wind speed position, amplitude, and timing agreed within 5%[reported], with $t_{\max}$ of 590 s and 600 s respectively[reported], but the stated agreement was limited to approximately one minute before to three minutes after the peak. Nguyen et al. (2018) found that in full-scale cold-pool parameterisation a modified Lundgren parameter can organise some cold-pool downdraft and outflow-front trajectories[reported]. These results indicate that the choice of event window directly affects the comparable time-span range.

Table 2: CM1/WRF source-field output variable contract. M = minimum required, P = recommended, S = solver-specific. Availability: ✓ = reported, ? = not reported.

Variable	Tier	CM1	WRF	Description and QA check
u, v, w (3-D velocity)	M	✓	✓	Staggered-grid position must be explicit; check for missing values
Timestamp sequence	M	✓	✓	Absolute or event-relative time; Δt_{out} must be recorded
Grid metric and coordinates	M	✓	✓	Horizontal spacing, vertical-layer definition, terrain/surface elevation
Coordinate-system definition	M	✓	✓	Source-coordinate origin, rotation, projection parameters
Terrain/surface elevation	M	✓	✓	Used together with vertical-layer semantics
Vertical-layer semantics	M	✓	✓	Height/mass/flux layer; σ or z coordinate
Source identification and provenance	M	✓	✓	Model version, configuration, initial-condition identifier
p or p' (pressure/perturbation pressure)	P	✓	✓	Full vs. perturbation must be distinguished; reference state must be reported
θ (potential temperature)	P	✓	✓	Required for thermodynamic reconstruction
T (temperature)	P	?	✓	CM1 typically derives from θ and p
Moisture variables (q_v, etc.)	P	✓	✓	Density reconstruction and buoyancy calculation
ρ or derivable density	P	?	?	If not directly output, reconstruction requires thermodynamic variables
TKE or SGS variables	S	?	?	Downstream solver inflow-turbulence initialisation
Staggering metadata	S	✓	✓	Arakawa C/B grid staggering-position specification
Surface fields (fluxes, etc.)	S	?	?	Ground boundary-condition matching
Restart identifier and checksum	S	?	?	One checksum per event subdomain per snapshot
Boundary-condition provenance	S	?	?	Lateral-boundary and top-driving source record

The variable contract transforms source-field evidence into deliverable data objects, but the

Table 3: Source-field variable to CFD boundary-variable transformation contract. Each row defines the input variable, target variable, required transformation, units, and QA check.

Source variable	**Target CFD variable**	**Transformation**	**Units**	**QA check**
u_s, v_s, w_s	U, V, W (target frame)	Storm-motion removal + coordinate rotation	m/s	Rotation-matrix determinant = 1; component conservation
p'_s or p_s	p_{CFD}	Reference-state reconstruction + pressure reference	Pa	p_{ref} explicitly defined; full/pertu distinction
θ_s	T_{CFD}	$T = \theta(p/p_0)^{R_d/c_p}$	K	Thermod consistency check
$q_{v,s}$	ρ_{CFD}	Equation of state + moist-air density	kg/m^3	Mass-flux residual < threshold
Staggered u point	Cell centre	Destaggering (linear or higher-order)	—	Momentu flux difference before and after destaggering
Source grid	Target grid	Trilinear/higher-order spatial interpolation	—	Interpola weights saved; boundary face registration

coordinate position, units, and reference state of the same variable may still differ between the two models. The downstream preparation process must make these differences explicit and retain transformation records sufficient for reverse checking.

3 Source-field preparation

Source-field preparation is not a single file-format conversion but the re-expression of meteorological-model semantics into data objects that a building-scale solver can use and that are reversibly auditable. The output package, beyond boundary and initial fields, should include coordinates, reference states, interpolation weights, flux residuals, and checksums, so that errors at each step can be tracked separately from source-field errors.

3.1 Coordinate system and reference frame

The transformation between the source-field coordinate system (typically the CM1 model grid or a WRF projection coordinate) and the building-centred target coordinate system must be decomposed into three independent operations: coordinate rotation (aligning source-field horizontal directions to the building reference direction), storm-motion removal or retention (depending on the chosen forcing representation), and physical translation (moving the origin to the building base centre). Coordinate rotation is described by an orthogonal matrix $\mathbf{R}$ whose determinant is identically 1. Storm-motion removal means subtracting the storm translation velocity $\mathbf{c}_s$ from the velocity field so that the event appears in the target coordinate system as a transient field passing over a fixed building; retaining storm motion means the building sees a different effective velocity in the moving reference frame. These two choices correspond to different boundary-condition role assignments and cannot be mixed within the same simulation.

Dang et al. (2026) provided the most complete interface record: CM1 u, v, w, and pressure perturbation were transferred into Fluent via ground-relative reference-frame conversion, trilinear spatial plus linear temporal interpolation[reported], from a 112 km domain, 10 m spacing, 0.2 s output step to 0.02 s LES step[reported]. That source explicitly documented the reference-frame conversion, but the physical event was a tornado rather than a downburst. Dang et al. (2023) transferred the velocity field from a CM1 10 m nested domain to a CFD domain via three-dimensional and temporal linear interpolation in one direction[reported], producing an asymmetric tangential velocity field and a tilted moving vortex, but the complete variable contract and output frequency were both [not reported].

3.2 Destaggering and interpolation

CM1 uses an Arakawa C-type staggered grid with velocity components u, v, w defined at face centres and scalars at cell centres. Transfer to a non-staggered CFD grid requires a destaggering operation (linear or higher-order), and the momentum-flux difference before and after destaggering must be recorded. Spatial interpolation maps source-grid values to the target CFD grid; trilinear interpolation is the most common choice, but higher-order interpolation can reduce numerical dissipation. Temporal interpolation establishes continuity between the source-field output time step Δt_{out} and the CFD time step Δt_{CFD}.

Dang et al. (2026)'s 10:1 temporal scale ratio (0.2 s source to 0.02 s LES)[derived] indicates that boundary temporal interpolation carries the discrepancy between source-field output and building-

scale stability time step. Nguyen (2021) already showed at the source-field level that when peak amplitudes are close, peak radii can still differ by 22.2% to 32.0%[reported]; if the interface further smooths moving structures at coarse time intervals, a secondary shift occurs in the building's position relative to the gust front.

3.3 Thermodynamic reconstruction and conservation

When recommended-contract variables are available, pressure and density can be reconstructed via the equation of state: $T = \theta(p/p_0)^{R_d/c_p}$, $\rho = p/(R_d T_v)$, where T_v is virtual temperature. After reconstruction, mass-flux and momentum-flux residuals must be checked against a predetermined threshold. If the source field provides only pressure perturbation p', the full pressure $p = p_0 + p'$ requires the explicit definition of the reference state p_0.

3.4 Compression and data management

High-frequency three-dimensional output generates large data volumes. Orf et al. (2019) evaluated ZFP lossy compression tolerances[reported], providing a reference for data pipelines of high-frequency CM1 output. Compression should be treated as a measured I/O option rather than a neutral storage step. Compression error must be quantified on velocity, pressure, spectral, and event-position observables before use. A reproducible interface requires saving not only the converted inlet files but also source-field timestamps, solver time steps, the spatial coordinate transformation matrix $\mathbf{R}$, interpolation weights, boundary role assignments, and mass corrections.

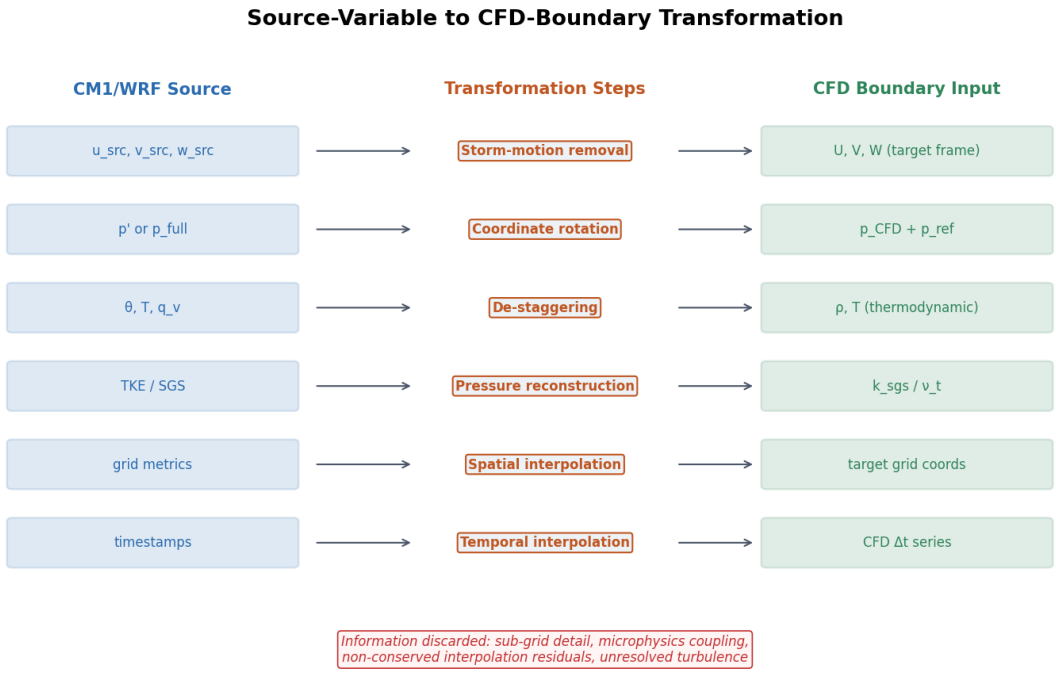


Figure 2: Transformation flow from source-field variables to CFD boundary variables. Left column: CM1/WRF source variables. Centre column: transformation steps. Right column: CFD boundary inputs. Bottom annotations indicate sources of information loss. This figure is a project-original synthesis diagram and does not constitute validation evidence.

After the above preparation, whether data can enter the fine-scale domain stably depends on

how much three-dimensional structure the boundary faces retain, how inflow/outflow role changes are handled, and in what manner unresolved turbulence is supplemented. The following section evaluates these choices along uniform comparison axes, avoiding the substitution of software names for interface definitions.

4 Constructing fine-scale CFD boundary and initial conditions

Boundary and initial conditions determine to what extent the three-dimensional non-stationary structure from the source field is preserved. Six strategies are evaluated on the same axes: required variables, spatial and temporal interpolation, conservation treatment, unresolved turbulence, boundary role, numerical stability, implementation burden, evidence type, and failure mode (Table 4), thereby placing information fidelity and implementation cost within the same framework.

4.1 Offline multi-face playback

Offline multi-face playback applies pre-converted source-field data via file interface to multiple boundary faces (lateral and top) of the CFD domain, preserving the three-dimensional spatial structure of the source field. Heldens et al. (2021) implemented the INIFOR/PALM system's offline dynamic interface, including coordinate rotation, time-varying initial and boundary fields, top global mass correction, and Xie–Castro synthetic turbulence generation[reported], benchmarked on a semi-idealised case but without observational verification. Salim et al. (2020) supplied all lateral boundaries and the top of an OpenFOAM URANS domain with time-varying three-dimensional velocity, TKE, thermodynamic, and surface fields from COSMO+DCEP, and applied global mass-flux correction[reported], but the two-day single-station comparison was only a preliminary assessment. Global mass correction cannot eliminate local boundary imbalances, STG cannot fully reproduce spectra and higher-order moments, and one-hour coupling may retain mesoscale artefacts.

For the fixed target, offline multi-face playback preserves the most source-field spatial information but requires complete three-dimensional velocity, pressure, and thermodynamic variables together with explicit flux matching at each face. The retained corpus contains no downburst-to-high-rise-building offline multi-face playback case.

4.2 Time-varying initial field plus lateral and top driving

Time-varying initial field plus lateral and top boundary driving provides a complete volume field at the initial instant and subsequently updates boundaries at a fixed frequency. Kadasch et al. (2021)'s WRF4PALM converted WRF output to PALM dynamic driving, with lateral boundary and top updates every hour[reported], using STG separately because the driving contained no resolved turbulence. Todea et al. (2012) supplied EULAG with WRF initial volume field and non-stationary lateral boundaries at 5 min frequency[reported], with the validation object being Ju2003 SF6 dispersion rather than structural loads. Update frequency in existing analogues spans a wide range: 0.2 s (Dang et al., 2026) to hourly (Heldens et al., 2021; Kadasch et al., 2021; Kumar et al., 2021; Sun et al., 2019).

4.3 Single-face profile transfer

Single-face profile transfer applies velocity and turbulence profiles on a single inlet face only. Yim et al. (2018) transferred WRF time-averaged profiles into complex-terrain steady RANS[reported]. Chen et al. (2022) fitted WRF static U, V, and temperature profiles separately on multiple Fluent inlet faces and superimposed WAWS turbulence[reported]. Single-face transfer is a lower-readiness option because it discards multi-face spatial structure; for a moving multi-directional downburst the information loss is greatest.

4.4 Dynamic multi-face inflow/outflow handling

When a downburst gust front translates, the inflow/outflow role of each CFD domain face switches over time. Kumar et al. (2021) read the hourly WRF wind direction through a Linux Shell coupler, reassigned OpenFOAM inlet and outlet faces, applied uniform inlet velocity, and ran a new steady RANS[reported]. This is an explicit boundary-role switch, but it replaces continuous transient evolution with repeated steady RANS and does not preserve flow memory during the event passage.

For the fixed target, dynamic role switching is conceptually closest to the time-varying multi-directional nature of downbursts, but the existing evidence contains no role-switching implementation at transient-field resolution.

4.5 Same-solver nesting or IBM

Same-solver nesting or immersed boundary methods (IBM) handle multiscale and building geometry within a single framework. Wiersema et al. (2020) pushed WRF dynamic nesting to a 2 m IBM urban domain, reporting horizontal wind-speed FAC2 of 0.91 (for PWIDS) and a minimum of 0.73 (for super-PWIDS) in the Joint Urban 2003 validation[reported]. Rodriguez et al. (2025) used a moving top disc and cyclicACMI in OpenFOAM to implement a moving downburst passing over fixed urban and terrain geometry[reported], with observational comparison limited to Genoa lidar vertical radial-velocity profiles at three instants. The same-solver route eliminates one file-interface fracture but introduces resolution-jump and wall-pressure risks.

4.6 Synthetic or recycled turbulence reconstruction

When the transferred mean flow does not contain resolved turbulence, high-wavenumber content must be reconstructed at the interface. Li et al. (2018)'s New-CP combined trilinearly interpolated WRF mean flow and potential temperature with database small-scale turbulence on a fixed LES inlet face[reported]. Natarajan and Hangan (2019)'s procedure refined meteorological velocity and potential temperature, reconstructing lost high-wavenumber turbulence via scale similarity plus a Smagorinsky method in an intermediate buffer region[reported]. Kondo et al. (2012) combined per-minute updated WRF mean velocity with LES-recycled fluctuations in a driver domain[reported].

All of the above analogues validated urban wind, dispersion, or typhoon conditions; none validated the error from convective-downburst forcing to building pressure. Applying any of these schemes to the downburst-to-high-rise-building chain requires fresh conservation, spectral, and pressure verification under downburst input conditions. Cao et al. (2011)'s review classified coupling approaches[reported]; Garcia-Santiago et al. (2018) used WRF ensemble time histories to define input distributions for steady RANS profiles and propagated uncertainty[reported]; Nakamura

et al. (2019) non-dimensionalised independent LES results in post-processing using single-point WRF wind speed[reported]. Yaghoobian et al. (2021)'s WRF–OpenFOAM–EnergyPlus framework transferred meteorological boundaries[reported]; Liu et al. (2018)'s AHW-to-WRF one-way nesting produced 148 m WRF-LES wind fields[reported]; Kwak et al. (2015) transferred meteorological and chemical profiles to RANS CFD at 30–60 min frequency[reported]; Ma et al. (2018) fitted WRF mean wind speed and turbulence intensity profiles at a single point[reported]; Wang et al. (2023) supplied three power-law parameters from a coastal grid point to a CityFFD inlet face every 20 min[reported]. These sources cover a wide range of applications from urban microclimate to bridge typhoon loading but none constitutes boundary-driving evidence for downburst to H=100 m target.

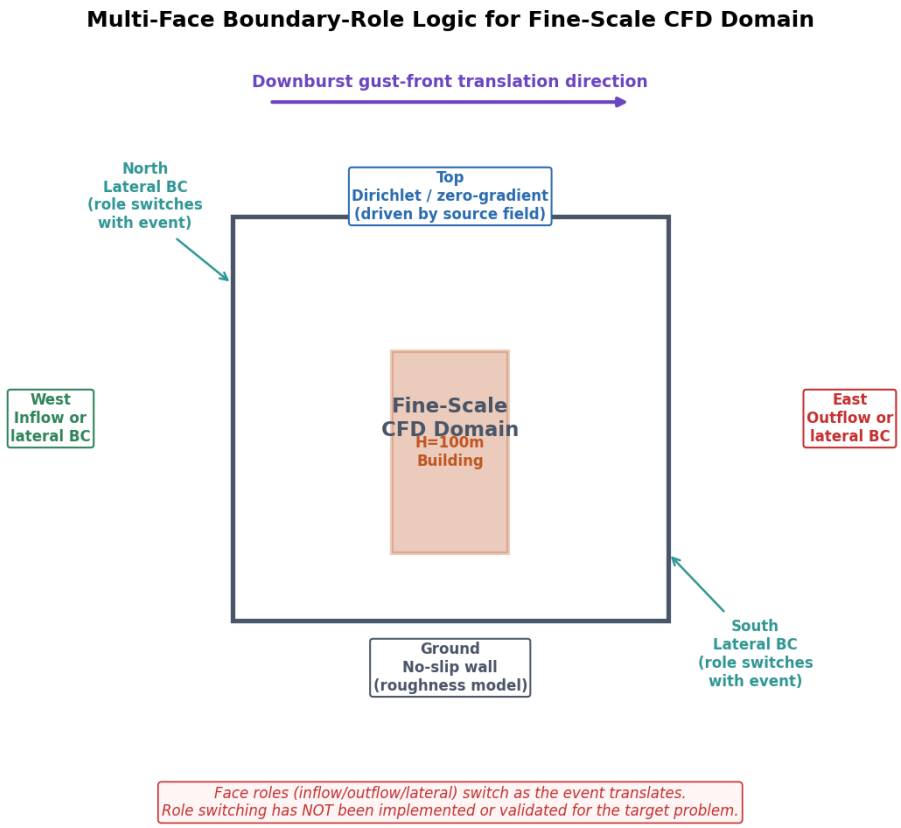


Figure 3: Schematic of multi-face boundary-role logic for the fine-scale CFD domain. Face roles switch as the downburst event translates. This role switching has not been implemented or validated for the target problem. This figure is a project-original synthesis diagram and does not constitute validation evidence.

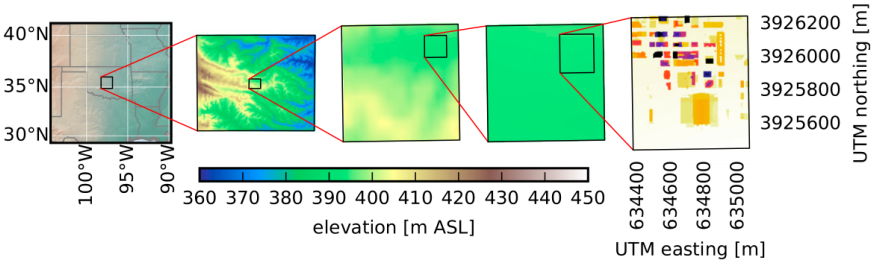


FIG. 6. Configuration of domains used in the multiscale simulation centered over the business district of Oklahoma City. The five domains have resolutions of 6.05 km and 550, 50, 10, and 2 m. The 550-, 50-, and 10-m domains include contour levels of topography. The 2-m domain includes contours of the building heights AGL (the color bar is not shown). Dimensions of each domain and other configuration information are included in Table 1.

maintaining an optimal grid aspect ratio near the surface in the region of interest.

c. Urban geometry

Urban geometry represented by the immersed boundary was created by sampling, at each grid point, a "shapefile" containing vectorized building information. Narrow gaps between buildings and other insufficiently resolved features were manually adjusted to ensure that each of the VRM interpolation points had a minimum of two nearest neighbors for the IDW interpolation. WRF-IBM currently uses a two-dimensional array to store the immersed boundary height at each grid point, which restricts the model topography to solid shapes without void space. Because of this restriction, an elevated walkway at UTM coordinates (634850, 3925800) was omitted from the model topography because WRF-IBM is currently unable to resolve flow in the free space beneath the suspended structure. In addition, several buildings near inflow boundaries, specifically the southern edge, were removed because of spurious interactions with the inflow conditions. Nonphysical behavior around buildings near the inflow boundaries of the 2-m domain is not unexpected because these buildings are not represented on the 10-m parent domain and the flow on the parent domain is unobstructed. Identical building geometry representations were used for both the idealized and multiscale simulations.

Variations in the ground elevation within the microscale modeling domain are small in magnitude, with minimum and maximum elevations of 360 and 365 m MSL. Because of restrictions from periodic boundary conditions, the idealized simulations ignore the underlying ground topography and only building heights

TABLE 1. Multiscale model configuration for JU2003 simulations for domains 1–5 (TF = terrain-following coordinate, KF = Kain–Fritsch cumulus parameterization, Smag = 3D Smagorinsky turbulence closure, WSM3 = WRF single-moment 3-class, RRTM = Rapid Radiative Transfer Model, and MM5 = Fifth-generation Pennsylvania State University–National Center for Atmospheric Research Mesoscale Model).

	D1	D2	D3	D4	D5
Δx and Δy (m)	6050	550	50	10	2
Coordinate	TF	TF	TF	IBM	IBM
Time step (s)	30	3	0.25	0.05	0.01
East–west grid points	221	232	254	241	351
South–north grid points	221	232	254	241	401
Bottom–top grid points	51	51	76	146	243
Turbulence	MYJ	MYJ	Smag	Smag	Smag
Microphysics	WSM3	WSM3	WSM3	None	None
Longwave radiation	RRTM	RRTM	RRTM	None	None
Shortwave radiation	Dudhia	Dudhia	Dudhia	None	None
Surface layer scheme	MM5	MM5	MM5	None	None
Land surface model	Noah	Noah	Noah	None	None
Cumulus parameterization	KF	KF	None	None	None

Figure 4: Internal-review excerpt, reproduced from Wiersema et al. (2020), original Fig. 6 and Table 1, physical PDF page 10. WRF five-level dynamic nesting from 6 km to 2 m IBM domain grid and time-step configuration. Copyright and permissions must be verified before external publication.

These strategies do not form a directly applicable ranking. For the present target, a boundary scheme should first pass a no-building domain flux, spectral, and pressure stability check; only after the interface error has been quantified can comparisons of grid, turbulence closure, and near-wall treatment be attributable.

5 Fine-scale wind-field CFD

The fine-scale solution stage inherits boundary errors and additionally introduces grid, turbulence closure, near-wall treatment, and statistical-duration errors. The solver name itself cannot delineate result credibility; the evaluation object should return to the target observable and its convergence condition.

Table 4: Comparison of fine-scale CFD boundary and initial-condition construction strategies. Six approaches evaluated on the same comparison axes.

Strategy	**Required variables**	**Conservation treatment**	**Turbulence treatment**	**Implementation burden**	**Evidence sources and type**
Offline multi-face playback	Full 3-D u, v, w, p, thermodynamics	Explicit flux matching required	Source-resolved + optional STG	High	Heldens et al. (2021) dynamic driving; Salim et al. (2020) all lateral + top
Time-varying initial field + lateral/top driving	Initial volume field + boundary updates	Global mass correction	STG or database reconstruction	Medium–high	Kadasch et al. (2021) WRF4PALM; Todea et al. (2012) EULAG 5 min update
Single-face profile transfer	Inlet-face velocity + turbulence profile	No multi-face constraint	Synthetic turbulence required	Low–medium	Yim et al. (2018) steady RANS; Chen et al. (2022) static profile + WAWS
Dynamic multi-face inflow/outflow	Full 3-D + role-switching logic	Local flux balance required	Same level as playback	High	Kumar et al. (2021) hourly switching; not reported
Same-solver nesting/IBM	Grid nesting or IBM geometry	Solver-internal conservation	Internal resolution jump	High	Wiersema et al. (2020) WRF-IBM 2 m domain; Rodriguez et al. (2025) cyclicACMI
Synthetic/recycled turbulence reconstruction	Mean profiles + statistical constraints	Not applicable	STG/recycling/database	Medium–high	Li et al. (2018) database + interpolation; Natarajan and Hangan (2019) scale-similarity; Kondo et al. (2012) recycling

5.1 Solver-neutral requirements

Solver-neutral requirements for fine-scale wind-field CFD precede back-end selection. The target domain must be defined relative to the moving event and the building, not copied from an ABL benchmark. Upstream, downstream, lateral, and top extents are requirements that need testing rather than universally prescribable values. Grid design must separate four resolution requirements: source-boundary resolution (matching interface interpolation accuracy), wake and shear-layer resolution (capturing separation and reattachment), facade-pressure resolution (25-band discretisation sufficient to extract storey forces), and near-wall treatment (choice of wall-resolved or wall-modelled). Velocity convergence is not pressure-peak or moment convergence.

5.2 Solver and configuration evidence from retained sources

Iida and Uematsu (2019) used LES to simulate pulsed and translating jets acting on a low-rise building, finding that including time-varying static pressure can change local C_p by approximately 0 to +0.8 (pulsed) and approximately 0 to −0.4 (translating)[reported]. Transient static pressure is part of the load definition and can materially change local pressure coefficients in a translating-downburst simulation. Yan et al. (2022) tested a high-rise building model under impinging-jet downburst conditions with different terrain roughness, finding the highest mean building pressure near r/D=1.0[reported] and that roughness reduced the maximum radial velocity and raised the height of its occurrence[reported].

Jog (2013) used Star-CCM+ steady RANS to simulate the interaction of an impinging-jet downburst with a cube, with approximately 0.9 M and 2.4 M cells for flow-field and structural cases respectively[reported]. Larger h/D produced stronger and more extensive near-ground flow[reported]. Steady RANS, cubic geometry, and scale make this unable to represent the target's transient peak pressures. Sim (2015)'s doctoral thesis contains multi-chapter downburst and ABL building CFD: Chapter 3 and Chapter 7 report opposite planar-density branches that the source did not reconcile[reported]; Chapter 8's stated approximately 9.8% deviation did not report the metric, denominator, or comparison basis[not reported]; Chapter 10's SAS Courant-number study showed velocity-profile agreement but loads still exhibited sensitivity[reported]. Different chapters used different configurations and evaluation systems and cannot be merged into a single validation judgement.

Rodriguez et al. (2025) used a moving top disc and cyclicACMI in OpenFOAM to implement a moving downburst passing over fixed urban geometry[reported], with observational comparison limited to lidar velocity profiles at three instants; pressure coefficients and aerodynamic loads were not reported.

5.3 ABL high-rise LES validation evidence

ABL high-rise LES is closest to the target in geometry and output completeness, but the wind-field physics are not equivalent. Thordal et al. (2020)'s CAARC benchmark reported mean-pressure R^2>0.97[reported], DWT/EWT base-moment discrepancies of M_x 10.5%, M_y 13.0%, M_z 12.8%[reported], and B90 torsion overestimate of 16.2%[reported]. Xing et al. (2022)'s Torre Gioia 22 model reported mean C_p R^2 of 0.90–0.96[reported] and fluctuating C_p R^2 of 0.70–0.92[reported] across six wind directions. Papp et al. (2021)'s GPU LES achieved mean-pressure correlation R=0.90, FAC2 0.92, and AAD 0.150 on an approximately 9 M-cell cube[reported], with fluctuating

correlation only R=0.73[reported]. Ricci et al. (2018) reported mean-pressure deviation below approximately 10%[reported] and RMS pressure deviation reaching 19%[reported].

Geleta and Bitsuamlak (2022)'s low-rise building case had peak C_p within ±10% of the BLWT[reported], but the LES optimal normalisation duration was an order of magnitude shorter than the corresponding BLWT duration[reported]. The 168 M-cell virtual wind tunnel used 256 cores for approximately 48 d net runtime[reported], and 9.6–21.8 M main domains used 64–160 cores for approximately 45–53 d net runtime[reported]. Melaku and Bitsuamlak (2024)'s three-stage CAARC ABL LES validation showed encouraging agreement but reported no universal summary error metric[reported]. Ciarlatani et al. (2023) used CharLES's low-Mach-number LES to resolve strong negative peak pressures at the leeward top corner of a high-rise building under oblique wind[reported], with the resolved scale in the critical region lying between the Taylor microscale and the Kolmogorov scale[reported].

High R^2 for mean pressure indicates that the mean-pressure field can be well reproduced under stationary ABL conditions, but one cannot infer equivalent accuracy under downburst conditions. Fluctuating accuracy is systematically lower than mean accuracy. Local peaks are the most difficult observable to converge: Xing et al. (2022) reported that the average fraction of local extremes falling within wind-tunnel Envelope 1 was 77.6% for maxima and 80.7% for minima[reported]; Envelope 2 reached 90.4% and 90.3%[reported].

5.4 Accelerating-flow cross-section CFD

Li et al. (2025) proposed the instantaneous non-dimensional acceleration parameter $a_p = D\dot{U}/U^2$ to organise accelerating-flow aerodynamic forces[reported]. In the selected sinusoidal acceleration cases, peak-drag exceedance from R3 to R16 increased from 22.4% to 109.7%[reported], the accelerating upstroke could be fitted by a Morison form with C_a=1.37[reported], and the downstroke exhibited hysteresis. The maximum drag deviation compared with wind-tunnel data was below 12%[reported]. The 2D URANS medium mesh comprised 180,000 cells[reported], the 3D LES approximately 3.6 M cells[reported], with mean y^+<2.1[reported]. Tan et al. (2026) used a realizable k-ε model with 490,400 cells, 10^{-4} s time step, maximum y^+=1.10, and maximum Courant number 0.94[reported], performing VMD and POD decomposition on a 3:2 cross-section at 10° angle of attack. The first two POD modes explained more than 92% of the energy[reported], and some covariance-mode correlation coefficients exceeded 0.98 or 0.99[reported]. These results support retaining the effect of the acceleration parameter in unsteady load analysis, but cross-section exceedance cannot be directly converted to full-storey pressure or a CM1 downburst amplification factor.

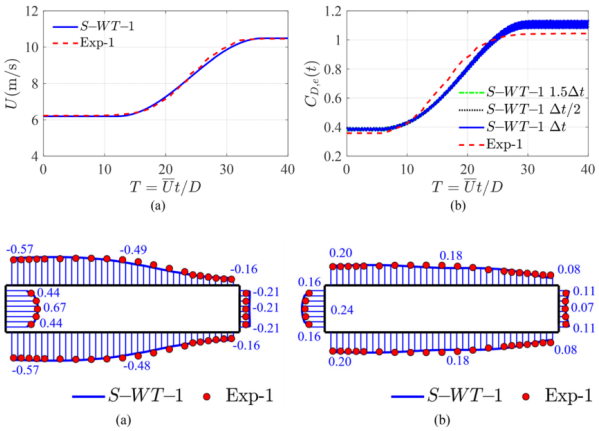


FIG. 5. Time step independence check under *S-WT*-1 conditions: (a) U and (b) $\overline{C}_{D,\sigma}(t)$.

FIG. 6. Pressure coefficients for *R5* under *S-WT*-1 conditions: (a) time-average and (b) standard deviation.

calculated using Eq. (10) show good agreement across different time steps and aligns well with the WTTs, with a maximum deviation of less than 12%.

Figure 6 further presents the comparison between the CFD and WTTs[56] in terms of pressure coefficients distribution of *R*5 under *S-WT*-1 during the acceleration phase. Overall, the mean and standard deviation of pressure coefficients obtained from present simulations are in good agreement with the WTTs.

Figure 7 shows the time-varying speed and drag coefficients of *R*3 under *S-WT*-2 inflows (see Table I) between CFD and WTTs.[22] It can be observed that CFD can well reproduce the target acceleration winds and the trend of the corresponding drag coefficients. This indicates that the present CFD setting can effectively simulate the aerodynamic characteristics of the rectangular cylinders under accelerating inflows.

V. RESULTS AND DISCUSSION

A. Accelerating inflows characteristics

Figure 8 shows URANS results for instantaneous velocity, non-dimensional parameter a_p and a_{ps} for the *Linear*, *Hyperbolic*, and *Sinusoidal*-type accelerating inflows. In particular, we remind that a_p is normalized by the instantaneous velocity using Eq. (2), and a_{ps} by the start velocity using Eq. (4). It can be found that CFD results match well with target theoretical values using Eqs. (5)–(7), respectively. As expected, for *Linear*-type inflow, a_p peaks at the acceleration start point and then ramp down rapidly, while a_{ps} stays constant. For the *Hyperbolic*-type inflow, a_p remains constant while a_{ps} ramp up with acceleration time. For *Sinusoidal*-type inflow, both a_p and a_{ps} first ramp up and then ramp down.

Coming now to the characterization of *Turbulent*-type accelerating inflow used in conjunction with LES, Fig. 9 present the time-history of the accelerating flow, while Fig. 10 presents the spectra of fluctuating turbulence components. Results at the inlet, 2.5*B*, and 5*B* from the inlet are in agreement with the target value, indicating that the synthetic turbulence is correctly transported in the LES domain.

B. Effects of acceleration parameters

Figure 11 provides the drag force for *R*4 under *Linear* and *Hyperbolic*-type accelerating inflows. In particular, the force

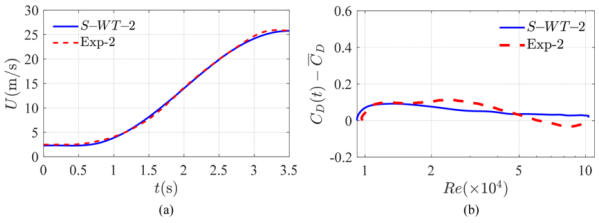


FIG. 7. Time-varying drag coefficients of *R*3 for acceleration case *S-WT*-2: (a) U and (b) $C_D(t) - \overline{C}_D$.

Figure 5: Internal-review excerpt, reproduced from Li et al. (2025), original Figs. 5–7, physical PDF page 8. Accelerating-flow time-step verification and CFD vs. wind-tunnel pressure/drag comparison. Copyright and permissions must be verified before external publication.

The existing evidence gives inconsistent accuracy for mean pressure, fluctuating pressure, and local peaks, indicating that fine-scale computation cannot end validation with a single velocity or mean-pressure metric. Before the pressure field enters load integration, the pressure reference, coordinate origin, face-element normals, and time basis must still be unified.

6 High-rise building pressure and load extraction

The reproducibility of building aerodynamic output depends on how dimensional pressure maps to element forces, facade-band forces, and base moments. Retaining dimensional quantities first and then computing coefficients only with a fully documented denominator semantics avoids non-stationary normalisation differences being mistaken for aerodynamic-force differences.

Table 5: Fine-scale CFD solver and configuration evidence comparison. All values are source-reported (marked reported) or not reported (marked not reported).

Source	Solver/method	Grid count	Δt	CFL	Wall treatment	Near-wall y^+
Iida and Uematsu (2019)	LES	Not reported	Not reported	Not reported	Not reported	Not reported
Yan et al. (2022)	Impinging-jet experiment	—	—	—	—	—
Jog (2013)	Star-CCM+ RANS	0.9–2.4 M	Not reported	Not reported	Not reported	Not reported
Sim (2015)	Multi-chapter, multi-config.	Not reported	Not reported	Not reported	Not reported	Not reported
Rodriguez et al. (2025)	OpenFOAM	Not reported	Not reported	Not reported	Not reported	Not reported
Thordal et al. (2020)	OpenFOAM LES	Not reported	Not reported	Not reported	Wall model	Not reported
Ricci et al. (2018)	OpenFOAM LES	Not reported	Not reported	Not reported	Wall model	Not reported
Papp et al. (2021)	GPU LES	∼9 M	Not reported	Not reported	Equidistant	Not reported
Xing et al. (2022)	OpenFOAM LES	Mesh M	Not reported	Not reported	Wall model	Not reported
Geleta and Bitsuamlak (2022)	OpenFOAM LES	9.6–168 M	Not reported	Not reported	Wall model	Not reported
Melaku and Bitsuamlak (2024)	OpenFOAM LES	Not reported	Not reported	Not reported	Wall model	Not reported
Ciarlatani et al. (2023)	CharLES LES	Not reported	Not reported	Not reported	Wall-resolved	∼10
Li et al. (2025)	2D URANS + 3D LES	180 k/3.6 M	Not reported	0.64–2.08	Wall-resolved	<2.1
Tan et al. (2026)	Realizable k-ε	490 k	10^{-4} s	0.94	Wall-resolved	1.10
Sabareesh et al. (2023)	LES/DES	Not reported	Not reported	Not reported	Not reported	Not reported

6.1 Coordinate system and sign convention

A right-hand building coordinate system is defined: x along the nominal along-wind direction, y in the cross-wind direction, z vertical. The origin is placed at the building base centre. M_x and M_y are overturning moments about horizontal axes; M_z is torsion about the vertical axis.

6.2 Dimensional pressure as the primary quantity

Dimensional pressure $p(t)$ is retained as the primary quantity. The element force is defined as

$$d\mathbf{F}(t) = -\big(p(t) - p_{\mathrm{ref}}(t)\big)\,\hat{n}\,dA \tag{1}$$

where p_{ref} is the pressure reference (fixed-event, time-varying, or height-based), $\hat{n}$ is the outward normal, and dA is the element area. The force on band i is

$$\mathbf{F}_i(t) = \int_{A_i} d\mathbf{F}(t) \tag{2}$$

Base shear is the sum over 25 bands:

$$\mathbf{V}_{\mathrm{base}}(t) = \sum_{i=1}^{25} \mathbf{F}_i(t) \tag{3}$$

The base moment is

$$\mathbf{M}(t) = \int_A \mathbf{r} \times d\mathbf{F}(t) \tag{4}$$

For the 25 facade bands, each band integrates only the horizontal resultant forces $F_{x,i}$ and $F_{y,i}$, excluding roof vertical forces. Following the right-hand coordinate system and $\mathbf{M} = \mathbf{r} \times \mathbf{F}$, the

specific expressions are

$$M_x(t) = -\sum_{i=1}^{25} F_{y,i}(t)\, z_i \tag{5}$$

$$M_y(t) = \sum_{i=1}^{25} F_{x,i}(t)\, z_i \tag{6}$$

$$M_z(t) = \sum_{i=1}^{25} \big(\Delta x_i F_{y,i}(t) - \Delta y_i F_{x,i}(t)\big) \tag{7}$$

where z_i is the height from the centre of band i to the base, and Δx_i and Δy_i are the horizontal offsets from the centre of band i to the reference point.

6.3 Pressure coefficients and normalisation

The pressure coefficient C_p must be accompanied by complete denominator semantics: pressure reference p_{ref}, density ρ, velocity reference U, reference height z_{ref}, and time dependence. Coefficients computed with a fixed-event reference ($q = \frac{1}{2}\rho U_{\max}^2$) and an instantaneous reference ($q(t) = \frac{1}{2}\rho U(t)^2$) are not interchangeable. Li et al. (2025) demonstrated in the accelerating-flow cross-section study that fixed terminal velocity and instantaneous denominator produce different transient interpretations[reported]. Iida and Uematsu (2019) reported that time-varying static pressure can change local C_p by approximately 0 to +0.8[reported]. If different faces use different references or denominators, the area-integrated forces and moments lack a common scale.

6.4 25-band discretisation and observable closure

The 25 facade bands convert the surface pressure field into an along-height distribution of storey forces. The pressure integration of each band requires retaining element area, normal, band number, reference pressure, and timestamp. The band count is a target definition, not a proven-sufficient resolution. Band-level convergence should check whether total base shear and moments are stable and also confirm that local peaks and peak timing are not smoothed by vertical aggregation.

Thordal et al. (2020) fully reported a storey-force-to-base $M_x/M_y/M_z$ plus torsion extraction architecture[reported]. Sabareesh et al. (2023) integrated surface static pressures into force and moment coefficients under a rigid-body assumption[reported]. Cao et al. (2019)'s tornado long-span dome response provided an alternative extreme-wind load-integration reference[reported]. These sources support the extraction architecture rather than target validation.

6.5 Error propagation from pressure to integrated loads

Pressure errors do not necessarily attenuate upon area integration. Thordal et al. (2020)'s mean-pressure R^2>0.97 coexists with base-moment discrepancies of 10.5% to 13.0%, and B90 torsion overestimate of 16.2%[reported]. Ricci et al. (2018)'s mean-pressure deviation below 10% and RMS pressure deviation reaching 19%[reported] further affect structural mapping. These results support a tiered error budget[bounded]: surface mean pressure, surface fluctuating pressure, local extremes, storey forces, base shear, two overturning moments, and torsion should each be given separate error figures; a prior-tier metric should not be used to infer that the next tier passes.

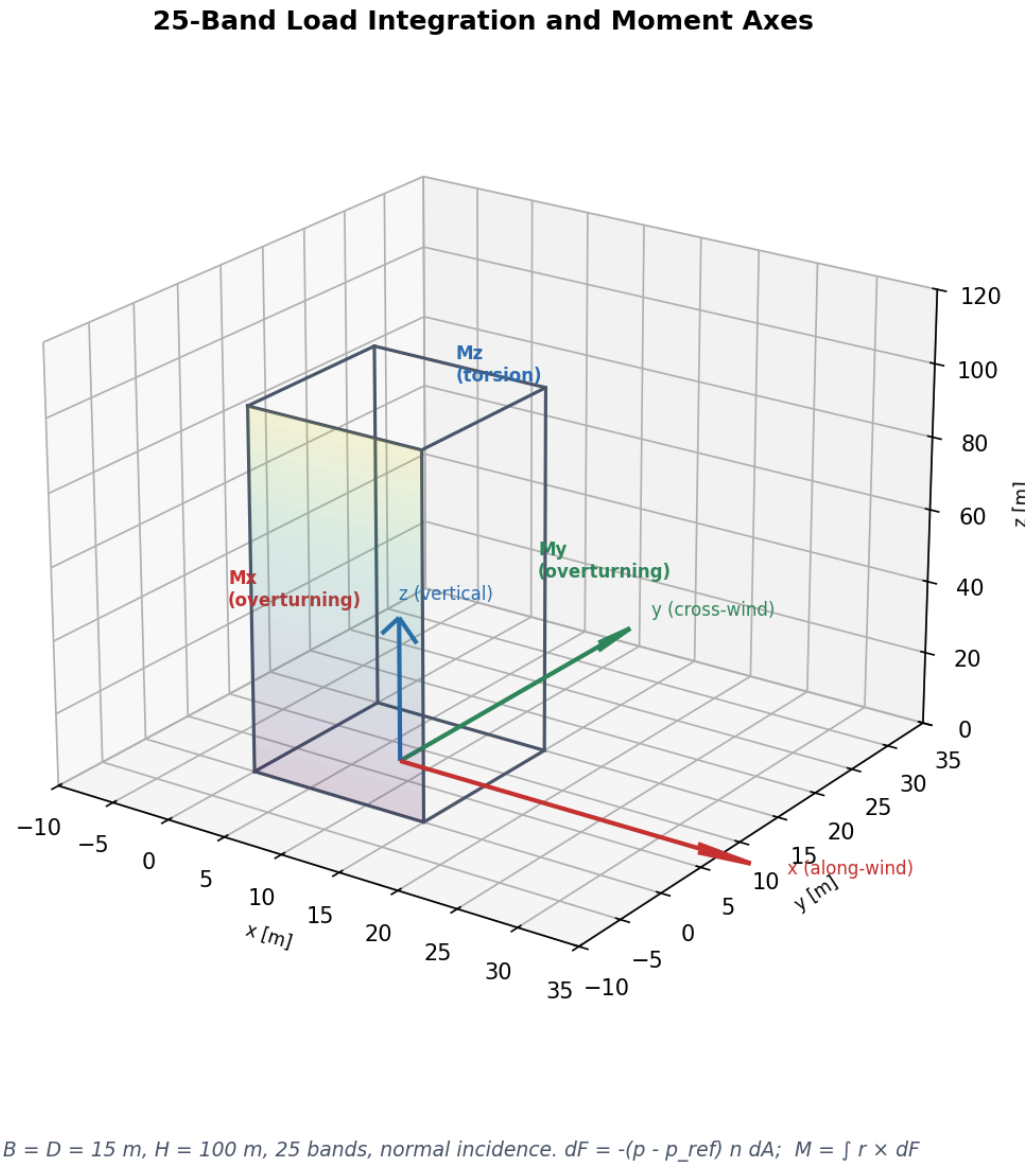


Figure 6: Schematic of 25-band load integration and moment-axis definition. $B{=}D{=}15\,\mathrm{m}$, $H{=}100\,\mathrm{m}$, 25 bands, normal incidence. M_x and M_y are overturning moments; M_z is torsion. This figure is a project-original synthesis diagram and does not constitute validation evidence.

The above definitions close the computational path from surface pressure to target aerodynamic endpoints but do not close the validation of those endpoints. Validation must proceed upward along the same data chain, retaining for each rung an independent comparison object and a failure record.

7 Validation ladder

Validation is divided along the data chain into eight progressive rungs (Table 7, Figure 7). R1 and R2 constrain the source field and interface, R3 constrains fine-scale velocity and spectra, R4 and R5 constrain facade pressure, and R6 through R8 constrain storey forces, base shear and overturning moments, and torsion respectively. Each rung requires independent inputs, comparison objects, metrics, and a pass-or-fail record; passing at a lower rung cannot substitute for evidence at a higher rung.

Table 6: Pressure and load definitions. All coefficients must carry complete denominator semantics.

Quantity	**Definition**	**Required accompanying information**
Raw pressure $p(t)$	Primary quantity, dimensional	Timestamp, element area dA, outward normal $\hat{n}$, element centroid
Pressure reference p_{ref}	Fixed-event/time-varying/height	Must be explicitly defined; fixed-event $\neq$ instantaneous normalisation
Pressure coefficient C_p	$(p - p_{\text{ref}})/q$	Density ρ, velocity reference U, reference height z_{ref}, time dependence of q
Element force $d\mathbf{F}$	$-(p - p_{\text{ref}})\hat{n}\, dA$	Element number, band membership, coordinates
Band force $\mathbf{F}_i$	$\int_{A_i} d\mathbf{F}$	Band i definition (height range, element-mapping rule)
Base shear $\mathbf{V}_{\text{base}}$	$\sum_{i=1}^{25} \mathbf{F}_i$	Sum of 25 bands; horizontal components V_x, V_y
Overturning moment M_x	$\sum F_{y,i}\, z_i$	About the base-centre origin; right-hand coordinate system
Overturning moment M_y	$\sum F_{x,i}\, z_i$	About the base-centre origin; right-hand coordinate system
Torsion M_z	$\sum(F_{x,i}\Delta y_i - F_{y,i}\Delta x_i)$	About the vertical axis; eccentricities Δx_i, Δy_i
Peak time series	$\max_t \lvert Q(t)\rvert$ and t^*	Statistical duration, filtering, sampling frequency
Spatial pressure correlation	$\rho_{pp}(r, \tau)$	Reference point, spacing, time lag, bandwidth

7.1 R1–R2: source field and interface

R1 checks whether source-field output maintains identity of peak position, amplitude, and timing before and after conversion. R2 checks interface mass-flux and momentum-flux residuals. These two rungs use direct timestep-by-timestep comparison of source-field data and converted data and do not involve the building. Nguyen (2021) already showed at the source-field level that when peak amplitudes are close, peak radii can still differ by 22.2% to 32.0%[reported].

7.2 R3: fine-scale velocity and spectra

R3 checks velocity profiles, TKE, and spectral slopes within the fine-scale CFD domain. Dang et al. (2026) provided velocity validation for a tornado terrain case, reporting that maximum horizontal wind speed at 10 m AGL increased by 29.5% relative to flat ground[reported]. Wiersema et al. (2020)'s WRF-IBM reported wind-speed FAC2 of 0.91 in Joint Urban 2003[reported]. Velocity agreement cannot substitute for pressure validation.

7.3 R4–R5: facade pressure

R4 evaluates mean and fluctuating facade pressure using metrics such as R^2, AAD, and FAC2. Thordal et al. (2020) reported mean-pressure R^2>0.97[reported]; Xing et al. (2022) reported mean-pressure R^2=0.90–0.96 and fluctuating R^2=0.70–0.92[reported]; Papp et al. (2021) reported mean-pressure R=0.90 and fluctuating R=0.73[reported]. R5 checks local pressure peaks; Xing et al. (2022) reported Envelope 1 hit rate 77.6%–80.7%[reported]. Ciarlatani et al. (2023) resolved the connection between corner negative peak pressure and multiscale vortex interaction[reported]. Pressure

agreement cannot be upgraded to force or moment agreement.

7.4 R6–R8: storey forces, base moments, and torsion

R6 compares band-force time histories; R7 compares base shear and overturning moment M_x, M_y deviations; R8 compares torsion M_z deviation. Thordal et al. (2020) reported DWT/EWT base-moment discrepancies of 10.5% to 13.0%[reported] and B90 torsion overestimate of 16.2%[reported]. Ricci et al. (2018) noted that mean integrated-force agreement did not uniformly extend to RMS internal forces[reported].

7.5 Applicability bounds of experimental validation evidence

Experiments enter the validation ladder primarily through 10 retained same-paper experiment-plus-CFD studies, each limited to its actually compared observable. The directed-validation edge VE-001 further records one restricted exception: Sim (2015) explicitly used the microburst-type high-rise experimental data of Zhang et al. (2014) to compare numerical results; that edge supports only pressure-coefficient and pressure-drag or viscous-drag comparison in an isolated-building or sparse-array context[bounded]. It does not validate moving downbursts, storey forces, base moments, torsion, other aspect ratios, general error bounds, or the end-to-end chain. Sengupta et al. (2008b), Sengupta et al. (2008a), Kim and Hangan (2013), and Zhang et al. (2013)'s experiment-plus-CFD comparisons likewise support only their actually reported observables.

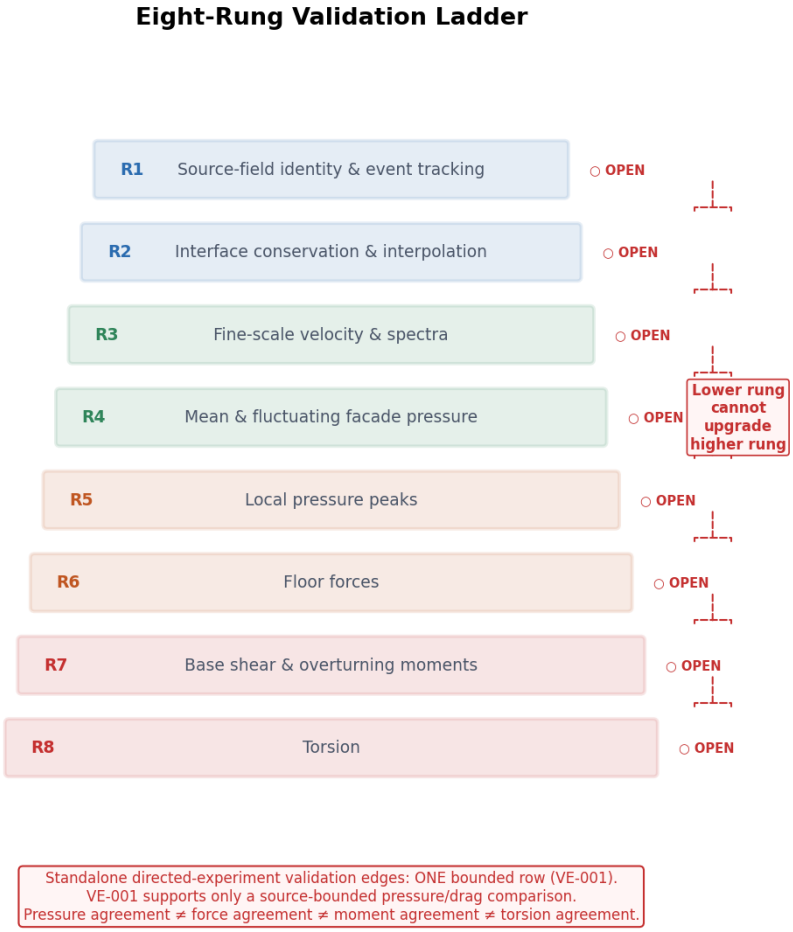


Figure 7: Eight-rung validation ladder and no-upgrade rules. All eight rungs are marked OPEN. The independent experimental directed-validation edge ledger contains one entry, supporting only source-limited pressure or drag comparison. This figure is a project-original synthesis diagram and does not constitute validation evidence.

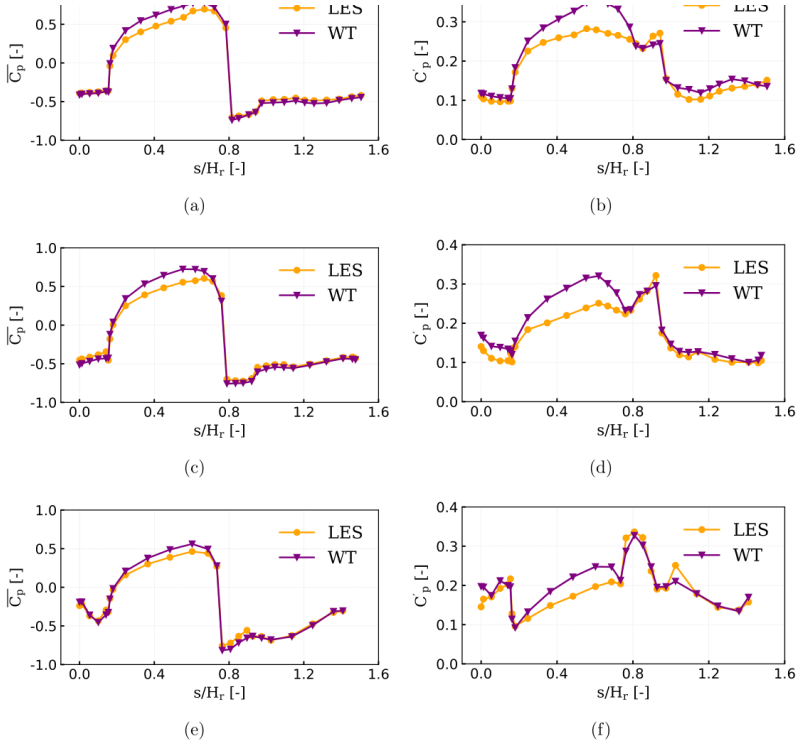


Fig. 14. Pressure coefficient statistics at 240° angle of attack: (a) $\bar{C}_p$ and (b) C_p' for Top path, (c) $\bar{C}_p$ and (d) C_p' for Medium path, (e) $\bar{C}_p$ and (f) C_p' for Bottom path.

Table 3
The percentage of extremes falling in the WT confidence interval.

Peak types		0°	60°	120°	180°	240°	300°	Mean
Maximum	Inside Env.1 ratio (%)	74.8	84.9	68.0	87.3	80.7	69.8	77.6
	Inside Env.2 ratio (%)	87.5	91.8	84.7	95.0	93.0	90.3	90.4
Minimum	Inside Env.1 ratio (%)	77.3	88.3	75.1	87.1	80.9	75.7	80.7
	Inside Env.2 ratio (%)	88.5	94.4	86.5	94.6	91.3	86.7	90.3

The proposed simulations have been run at CINECA, on the GALILEO supercomputer, which is equipped with 1022 nodes each composed of 2x18-core Intel Xeon E5-2697 v4 (Broadwell) at 2.30 GHz. Running each simulation with a number of cores comprised between 100 and 200, each simulation required approximately 5000 *Core-h* to complete for the *Mesh M* 10 min in real scale, which should be considered a minimum requirement in agreement, for instance, with CNR [34].

Although in the present study, for the sake of convenience, simulations have been performed every 60°, a stepping comprised between 10° and 30° is usually required [34]. Assuming for instance to sample the attack angle every 30°, this requires 12 simulations to complete the attack angles.

It is thus easily possible to see that a complete study performed as described in this paper requires approximately 6.0E4 *Core-h* if every angle of attack is sampled for 10 mins, while it would require 3.6E5 *Core-h* for a 1 *h* sample in full-scale, from which a more careful evaluation of the extreme values can be obtained.

According to Rescale [35],Walker [36], we here assume a cost of the *Core-h* equal to 0.1 euro (this might be considered an average value with typical oscillations of about 20%). This leads to a total cost comprised between 3k and 50k euro, which is indeed the range of price of wind tunnel tests.

It is nevertheless noticed that the cost structure of wind tunnel tests and numerical simulations is completely different: the fist ones are characterized by fixed costs due to the wind tunnel infrastructure and

Figure 8: Internal-review excerpt, reproduced from Xing et al. (2022), original Fig. 14 and Table 3 with Section 5.6, physical PDF page 10. Real high-rise local pressures, peak confidence-interval hit rates, and HPC cost. Copyright and permissions must be verified before external publication.

The validation ladder places the local evidence from different sources back within their respective observables, preventing model voting from forming a spurious end-to-end conclusion. Based on this boundary, the comparison of candidate routes shifts its focus to data contracts, executable verification, and interfaces that can be traced back upon failure.

8 Executable workflows for the present PETW problem

Three candidate routes share the same pressure and load endpoints but differ in source-field type, boundary expression, and implementation burden. Implementation order is determined by interface precedent and verifiability within the retained literature; it serves only to arrange the first short-duration verification and does not indicate that a solver has been selected or that any route has been executed or validated.

Table 7: Eight-rung validation ladder and experimental validation edges. Each rung has independent metrics; passing at a lower rung must not be upgraded to a higher rung. The independent experimental directed-validation edge ledger contains one entry.

Rung	Observable	Metric	Same-paper experiment edge	Prohibited inference
R1	Source-field identity and event tracking	Peak position, amplitude, timing match	—	Must not skip to R2
R2	Interface conservation and interpolation	Mass/momentum flux residual	—	Flux balance ≠ spectral fidelity
R3	Fine-scale velocity and spectra	Velocity profile, TKE, spectral slope	Dang et al. (2026) (tornado terrain velocity)	Velocity ≠ pressure
R4	Mean and fluctuating facade pressure	R^2, AAD, FAC2	Thordal et al. (2020) R^2>0.97; Xing et al. (2022) R^2=0.90–0.96	Mean pressure ≠ peak
R5	Local pressure peaks	Envelope hit rate, peak C_p	Xing et al. (2022) Env1 77.6%–80.7%; Ciarlatani et al. (2023) corner resolution	Peak ≠ storey force
R6	Storey forces	Band-force time-history comparison	Thordal et al. (2020) storey-force extraction	Storey force ≠ base moment
R7	Base shear and overturning moment	M_x, M_y deviation	Thordal et al. (2020) DWT/EWT discrepancy 10.5%–13.0%	Moment ≠ torsion
R8	Torsion	M_z deviation	Thordal et al. (2020) B90 overestimate 16.2%	Must not claim end-to-end
VE-001	High-rise microburst pressure or drag comparison	Source-limited benchmark	Sim (2015) using Zhang et al. (2014) experiment	Must not extrapolate to storey force, base moment, torsion, or end-to-end chain

8.1 Workflow A: CM1 output to engineering LES via offline multi-face playback

Workflow A can be prioritised for first interface verification. The retained corpus includes direct CM1-to-second-solver velocity-transfer precedent (Dang et al., 2023, 2026) and building-resolving CFD methods (Thordal et al., 2020; Xing et al., 2022; Papp et al., 2021; Ciarlatani et al., 2023), but the two categories of evidence have never been combined end to end on the same downburst event; the priority order therefore reflects only interface proximity.

Prerequisites are CM1 event output (u, v, w, p', θ together with coordinates and timestamps) and offline variable-conversion tools. The input package comprises three-dimensional velocity field plus pressure plus thermodynamic variables plus the coordinate transformation matrix $\mathbf{R}$ plus the timestamp sequence. The transformation sequence is storm-motion handling, coordinate rotation, destaggering, thermodynamic reconstruction, spatial interpolation, temporal interpolation, and mass correction. The solver-neutral configuration is a building-centred domain, partitioned grid (source-boundary / wake / facade / near-wall), LES or appropriate turbulence model, and 25-band output definition.

The QA-gate sequence is: (1) round-trip conversion check (source field → conversion → inverse conversion → comparison), (2) closed-domain flux balance (no building, closed domain, check mass and momentum flux residuals), (3) no-building dynamic-boundary stability test, (4) stationary-building pressure plausibility check (fixed position, C_p comparison under known ABL conditions). These are proposed tests, not completed runs.

Resource evidence[bounded]: Papp et al. (2021)'s single-GPU approximately 9 M-cell ABL case completed within 6 h[reported]; Xing et al. (2022)'s single-direction 10 min LES approximately 5,000 core-hours[reported], 12-direction approximately 60,000 core-hours[reported]. Additional costs for downburst event ensembles, interface sensitivity, and grid refinement are not included.

Unresolved risks are the influence of CM1 output frequency on building pressure, multi-face transient flux balance, and spectral completeness comparison between source-field turbulence and

interface-reconstructed turbulence.

8.2 Workflow B: WRF/WRF-LES to PALM/OpenFOAM/Fluent via dynamic boundary

Workflow B is an input-dependent alternative path, conditional on WRF products and dynamic driving variables being available. Kadasch et al. (2021)'s WRF4PALM[reported] and Heldens et al. (2021)'s INIFOR[reported] provide WRF-to-PALM dynamic driving interface frameworks. Salim et al. (2020) demonstrated COSMO-to-OpenFOAM full-boundary time-varying driving[reported]. Prerequisites are WRF dynamic driving variables (lateral and top velocity, TKE, thermodynamics) and STG or an equivalent synthetic turbulence tool. The QA-gate sequence is the same as Workflow A, plus dynamic driving variable completeness verification and STG spectral check.

Unresolved risks are whether the spatiotemporal resolution of WRF products is sufficient to drive downburst-scale building LES, and the fidelity of STG under non-stationary conditions.

8.3 Workflow C: same-solver nesting or IBM/overset handling

Workflow C is a higher-burden alternative path, suitable when a moving event or complex geometry needs to be handled within a single framework. Wiersema et al. (2020)'s WRF-IBM five-level nesting to 2 m domain[reported] and Rodriguez et al. (2025)'s OpenFOAM cyclicACMI[reported] provide technical precedent. Prerequisites are a same-solver version supporting nesting or IBM geometry. QA gates add nesting resolution-jump checking and IBM wall-pressure accuracy checking.

Unresolved risks are pressure artefacts at the resolution jump, IBM wall-flux accuracy, and time-varying geometry handling for a moving event.

8.4 First executable tests

All three routes should begin with short-duration verification rather than proceeding directly to long-time batch computation: (1) round-trip conversion, (2) closed-domain flux balance, (3) no-building dynamic-boundary stability, (4) stationary-building pressure plausibility. These steps delineate the next executable check and do not represent completed runs.

Table 8: Computational resource and I/O evidence. All values are bound to hardware, grid, physical duration, and output conditions.

Source	Grid count	Hardware	Physical duration	Wall-clock time	Output frequency	Accuracy metric
Papp et al. (2021)	∼9 M	Single GPU	20 s	<6 h	Not reported	Mean-C_p R=0.90, fluctuating R=0.73
Xing et al. (2022)	Mesh M	100–200 cores	10 min/direction	∼5000 core-h/direction	Not reported	Mean R^2=0.90–0.96
Geleta and Bitsuamlak (2022)	168 M (VWT)	256 cores	—	∼48 d (net)	Not reported	Peak C_p within ±10%
Geleta and Bitsuamlak (2022)	9.6–21.8 M	64–160 cores	—	∼45–53 d (net)	Not reported	Same as above
Orf et al. (2019)	$>10^{11}$	Blue Waters	Not reported	Not reported	0.2 s subdomain	Meteorological visualisation
Vermeire (2016)	147 M	Not reported	Not reported	Not reported	Not reported	Cold-pool parameter study
Mason et al. (2010)	Not reported	Not reported	Not reported	Not reported	Not reported	13-scenario parameterisation
Dang et al. (2026)	Not reported	Not reported	Not reported	Not reported	0.2 s/0.02 s	Terrain velocity +29.5%/−29.9%
Tian et al. (2026)	Not reported	Single GPU	—	∼1 min inference	—	Mean-velocity error <5%
Li et al. (2025)	180 k/3.6 M	AMD EPYC	Not reported	Not reported	Not reported	Drag deviation <12%

Table 9: Comparison of the three candidate executable workflows. Preference order is based on implementation readiness and does not constitute solver selection, completed execution, or design guidance.

Attribute	**Workflow A (priority verification candidate)**	**Workflow B**	**Workflow C**
Path	CM1 → offline multi-face playback → engineering LES	WRF/WRF-LES → dynamic driving → PALM/OpenFOAM/Fluent	Same-solver nesting or IBM/overset
Prerequisites	CM1 event output (u, v, w, p', θ); offline conversion tools	WRF dynamic driving variables; WRF4PALM or equivalent interface	Same-solver support for nesting or IBM geometry
Input package	3-D velocity + pressure + thermodynamics + coordinates + timestamps	WRF lateral/top driving + STG	Nested parent-domain field
Interface type	File-level offline multi-face	Dynamic driving (online or offline)	Solver-internal nesting
QA gates	Round-trip conversion check, closed-domain flux balance, no-building boundary stability, stationary-building pressure check	Same as A + dynamic driving variable verification	Nesting resolution-jump check + IBM wall-pressure check
Resource evidence	Papp et al. (2021) single GPU <6 h (9 M ABL); Xing et al. (2022) ∼5000 core-h/direction	Kadasch et al. (2021) WRF4PALM; Heldens et al. (2021) INIFOR	Wiersema et al. (2020) WRF-IBM 2 m; Rodriguez et al. (2025) cyclicACMI
Unresolved risks	CM1 output frequency, multi-face flux balance, turbulence completeness	WRF product availability, STG non-stationary fidelity	Resolution-jump pressure artefacts, moving-event geometry handling
First test	Round-trip conversion → closed-domain flux → no-building stability → stationary-building C_p	Same as A sequence + dynamic driving stability	Nested wind field → IBM/overset building C_p
Verification order	Priority for first interface verification	Conditional alternative (depends on WRF products)	Higher-burden alternative

9 Focused remaining research needs

Remaining research needs retain only those questions that would directly change the executability of the current computational chain or the fidelity of target observables. Their common criterion is whether a new method can narrow an identified interface error, not whether it possesses higher algorithmic complexity.

9.1 Unresolved turbulence reconstruction

When a mesoscale source field does not contain resolved turbulence, the core challenge for interface reconstruction is whether the higher-order moments, spatial coherence, and peak timing of a non-stationary downburst can be recovered by existing STG or recycling methods. Heldens et al. (2021) and Kadasch et al. (2021) used STG[reported]; Li et al. (2018) used database reconstruction[reported]; Natarajan and Hangan (2019) used scale similarity[reported]. Verification must extend from inlet spectra to corner peak pressure, facade correlation, and integrated moments.

9.2 Multi-face transient boundary robustness

Dynamic role switching is conceptually closest to the time-varying multi-directional nature of downbursts, but the retained corpus contains no role-switching implementation at transient-field resolution. Kumar et al. (2021)'s hourly steady switching[reported] does not preserve flow memory. Pressure fluctuations at boundary corners and along-wind-direction transitions are the primary failure mode.

9.3 Moving-event AMR/overset/IBM

Adaptive mesh refinement (AMR) can concentrate computational effort on the moving gust front and separation zones. Wiersema et al. (2020)'s WRF-IBM[reported] and Rodriguez et al. (2025)'s cyclicACMI[reported] provide technical precedent but did not report surface-pressure accuracy for a sharp-edged high-rise building. Wall-modelled LES, DES, and IDDES log-layer mismatch and corner-separation misjudgement need to be constrained jointly by wall-resolved LES and wind-tunnel pressure data.

9.4 GPU acceleration

Papp et al. (2021) kept an approximately 9 M-cell ABL LES within 6 h on a single GPU[reported], but the fluctuating correlation was only 0.73[reported]. The limitations of equidistant grids for edge gradients and fluctuating quantities require more flexible GPU grid schemes.

9.5 Output compression and reduced-order post-processing

Orf et al. (2019) evaluated ZFP lossy compression tolerances[reported]. Tian et al. (2026)'s FNO, trained on 30 real-city-geometry LES cases, achieved mean-velocity error below 5% after initialisation[reported] and reduced spin-up by 29.8% to 48.8%[reported]. The FNO initialisation training data covered only a single atmospheric condition[reported]; output is smoother and must be independently validated on out-of-distribution downburst inputs. Tan et al. (2026)'s VMD–POD[reported] and Garcia-Santiago et al. (2018)'s uncertainty propagation[reported] provide

reduced-order and data-constraint tools, but modal-energy concentration is not force or pressure prediction error.

Wang et al. (2021) summarised meso–microscale coupling obstacles from a review perspective[reported]; Tiwari et al. (2024) surveyed OpenFOAM CWE literature and noted that neutral-ABL numerical methods cannot automatically be considered applicable to transient downbursts[reported].

10 Concluding remarks

The 57 retained sources collectively point toward a first-verification route closest to present conditions: convert CM1 event output to an offline multi-face boundary package, then check velocity, pressure, and 25-band loads in a building-resolving domain. Direct CM1-to-second-solver transfer precedent (Dang et al., 2023, 2026) and building-resolving LES evidence (Thordal et al., 2020; Xing et al., 2022; Papp et al., 2021; Melaku and Bitsuamlak, 2024; Ciarlatani et al., 2023) respectively support the interface and downstream method but have not closed the physical and validation gap between them. Round-trip conversion, closed-domain flux balance, no-building dynamic-boundary stability, and stationary-building pressure plausibility form a sequential verification sequence. Resource evidence ranges from a single-GPU approximately 9 M-cell ABL case in 6 h (Papp et al., 2021) to multi-directional high-rise LES at approximately 60,000 core-hours (Xing et al., 2022), but these figures do not include downburst event ensembles, interface sensitivity, and grid-refinement costs.

End-to-end CM1/WRF downburst to H=100 m high-rise building aerodynamic load chain has not been validated. The variable contract is incomplete, the influence of interpolation on peak position has not been quantified, non-stationary turbulence reconstruction has not been validated, multi-observable grid convergence has not been simultaneously reported, integration amplification has not been checked under downburst conditions, and data-conversion and post-processing costs are typically omitted from end-to-end resource budgets. These six interface fractures prevent splicing link-level evidence into an end-to-end validation claim. Until these conditions are met by independent evidence on the same chain, the problem remains implementable but not end-to-end validated, and the existing corpus is insufficient to form engineering adoption, code provisions, solver selection, or final structural-response conclusions.

```
scientific_approval_status = not_scientifically_approved
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